WORKED SOLUTION

STEP I 2006, Question 13

Probability and Statistics • Poisson Mixtures

STEP I 2006, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q13

The key idea

Diamonds scattered independently at a constant average rate are a Poisson process, so the number of diamonds in m kilograms of chocolate is Poisson with mean m. The amount of chocolate is itself random, so we condition on it: work out the answer for a fixed amount, then average over the die rolls using the law of total probability (and its cousin for expectations). Each part then reduces to summing a series, finite in (i) and geometric in (ii).

Solution

Let D be the number of diamonds I find. Since the diamonds are hidden independently at an average rate of one per kilogram, the number in a bar of mass m kg has the Poisson distribution with mean m. In particular, given a mass of m kg,

$$\mathrm{P}(D = 0) = \mathrm{e}^{-m}, \qquad \mathrm{E}(D) = m.$$

Throughout, $100k$ grams is $0.1k$ kilograms.

Part (i)

Let N be the score on the die, so $\mathrm{P}(N = n) = \frac{1}{6}$ for $n = 1, \ldots, 6$. Given $N = n$, I buy $0.1n$ kg, so $D \mid N = n$ is Poisson with mean $0.1n$. By the law of total probability,

$$\begin{aligned}\mathrm{P}(D = 0) &= \sum_{n=1}^{6} \mathrm{P}(N = n)\,\mathrm{P}(D = 0 \mid N = n) = \frac{1}{6}\sum_{n=1}^{6} \mathrm{e}^{-0.1n} \\ &= \frac{1}{6}\left(\mathrm{e}^{-0.1} + \mathrm{e}^{-0.2} + \cdots + \mathrm{e}^{-0.6}\right).\end{aligned}$$

This is a geometric series with first term $\mathrm{e}^{-0.1}$, common ratio $\mathrm{e}^{-0.1}$ and six terms, so

$$\mathrm{P}(D = 0) = \frac{\mathrm{e}^{-0.1}}{6}\left(\frac{1 - \mathrm{e}^{-0.6}}{1 - \mathrm{e}^{-0.1}}\right),$$

as required.

For the expectation we condition in the same way:

$$\mathrm{E}(D) = \sum_{n=1}^{6} \mathrm{P}(N=n)\,\mathrm{E}(D \mid N=n) = \frac{1}{6}\sum_{n=1}^{6} 0.1n = \frac{0.1\times 21}{6} = 0.35.$$

In other words, $\mathrm{E}(D) = 0.1\,\mathrm{E}(N) = 0.1\times 3.5$, which is what we had to show.

Part (ii)

Now let T be the number of the throw on which the first 6 appears. The throws are independent, and $T=t$ means $t-1$ non-sixes followed by a six, so

$$\mathrm{P}(T=t) = \left(\tfrac{5}{6}\right)^{t-1}\tfrac{1}{6}, \qquad t = 1,2,3,\ldots$$

Given $T=t$, I buy $0.1t$ kg, so $\mathrm{P}(D=0 \mid T=t) = \mathrm{e}^{-0.1t}$. Hence

$$\mathrm{P}(D=0) = \sum_{t=1}^{\infty}\left(\tfrac{5}{6}\right)^{t-1}\tfrac{1}{6}\,\mathrm{e}^{-0.1t} = \frac{\mathrm{e}^{-0.1}}{6}\sum_{t=1}^{\infty}\left(\tfrac{5}{6}\mathrm{e}^{-0.1}\right)^{t-1}.$$

Ask yourself Why may we sum this to infinity?

It is a geometric series with ratio $r = \frac{5}{6}\mathrm{e}^{-0.1}$, and $0<r<1$, so it converges with sum $1/(1-r)$. Therefore

$$\mathrm{P}(D=0) = \frac{\mathrm{e}^{-0.1}}{6}\cdot\frac{1}{1-\frac{5}{6}\mathrm{e}^{-0.1}} = \frac{\mathrm{e}^{-0.1}}{6-5\mathrm{e}^{-0.1}},$$

multiplying top and bottom by 6. This is the required result.

For the expected number of diamonds,

$$\mathrm{E}(D) = \sum_{t=1}^{\infty}\mathrm{P}(T=t)\,\mathrm{E}(D\mid T=t) = \sum_{t=1}^{\infty}\left(\tfrac{5}{6}\right)^{t-1}\tfrac{1}{6}\times 0.1t = \frac{0.1}{6}\sum_{t=1}^{\infty} t\left(\tfrac{5}{6}\right)^{t-1}.$$

This is where the hint comes in. For $|x|<1$ the binomial expansion gives

$$(1-x)^{-2} = 1+2x+3x^2+4x^3+\cdots = \sum_{t=1}^{\infty} t x^{t-1}.$$

With $x=\frac{5}{6}$ the sum is $\left(1-\frac{5}{6}\right)^{-2} = 36$, so

$$\mathrm{E}(D) = \frac{0.1}{6}\times 36 = 0.6.$$

Again this is $0.1\,\mathrm{E}(T)$, where $\mathrm{E}(T)=6$ is the familiar mean number of throws needed to get a 6.

Checks

- Numerically, the probability in (i) is about 0.7150 and in (ii) about 0.6131. A simulation of 2×10^5 shoppers, with Poisson diamond counts, gave 0.7151 and 0.6136, and mean numbers of diamonds 0.350 and 0.599.
- The probability in (ii) is smaller than in (i). That makes sense: on average I buy 600 g rather than 350 g.
- Both expectations are 0.1 times the mean of the random multiplier. Expected diamonds equal expected kilograms, since the expected number of diamonds given the mass is linear in the mass.

Common slips

- Working out $\mathrm{e}^{-0.35}$ (or $\mathrm{e}^{-0.6}$) for the probability of no diamonds. That uses only the average mass, but e^{-m} is not linear in m, so the averaging must come after taking e^{-m}, not before. Expectations are linear, which is why the shortcut does work for $\mathrm{E}(D)$.
- Forgetting to convert grams to kilograms, so getting a mean of $100n$ rather than $0.1n$.
- Writing $\mathrm{P}(T = t) = (\frac{5}{6})^t \frac{1}{6}$, which counts one non-six too many.

Beyond STEP

Here the parameter of a Poisson distribution is itself random, which makes D a *mixture* distribution. The tools used, the law of total probability and the tower law $\mathrm{E}(D) = \mathrm{E}\big(\mathrm{E}(D \mid T)\big)$, are central in first-year probability. Probability generating functions handle such problems neatly: $\mathrm{P}(D = 0)$ is the generating function of T evaluated at $\mathrm{e}^{-0.1}$. Fixing a random number of trials and then a random count in each one leads on to compound distributions and branching processes. Grimmett and Welsh, *Probability: An Introduction*, develops all of these.

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